Impact of Shape and Size of the Network on the Performance Effectiveness of Source Location Privacy Preservation Techniques Developed for WSNs

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***Abstract*—Wireless Sensor Networks (WSNs) can be deployed in different forms. The two widely-used network models are square and circle. Most of the research works on Source Location Privacy (SLP) in WSN consider either one of the mentioned network models. In this research, we focus on evaluating the two network models—square and circular topologies. We use the existing random walk-based routing protocols namely, Phantom Routing-based L-path Random Walk (PRLPRW) and Phantom Routing-based Pure Random Walk (PRPRW) to evaluate the performance of the SLP protocols under these two deployment models. Particularly, we use the network area, the number of nodes and the radio range as parameters to study the performance. We jointly examine the concerns of source location privacy and the amount of energy utilized by the sensor network when creating and assessing privacy-aware routing schemes. The effectiveness is assessed using performance metrics such as Safety Period, Capture Percentage, Entropy, Delay, Energy Consumption, and Network lifetime that have been developed, and simulation results show that the circular deployment model performs better.**

***Index Terms*—Source Location Privacy, WSN, Network Shape and Size, Radio Range, Performance Effectiveness, IoT.**

I. Introduction

Wireless Sensor Networks (WSNs) and their applications seem to be the subject of numerous studies in the past twenty years. Several Wireless Sensor Networks (WSNs) and their variants such as the Internet of Things (IoT) and Industrial Internet of Things (IIoT) have been a high priority because of the availability of low-cost, short-range sensors and their ease of deployment. WSN systems are typically designed to detect, track and transmit real-time data from specified monitoring settings to back-end systems for additional processing and analysis. WSNs have seen themselves in numerous applications such as health care, agriculture, as a predecessor for IoT, home automation, automotive networks, and Industry 4.0; etc., [1]–[6]. Recently, WSNs have been used in landslide detection monitoring, underwater applications, military, and wildlife protection [7]–[10] etc. In wildlife applications, WSNs are used to track the endangered species and provide updates on the whereabouts and health status of these animals [11], [12]. Since these networks are located at remote locations, their applications nevertheless face substantial obstacles such as network security risks, data collection, network deployment, network coverage and Source Location Privacy (SLP), etc [13]–[16]. Once such application that has attracted researchers across the globe is source location privacy; this is the domain our team has been exploring in this paper. Sensor node devices' comparatively simple functioning makes them easier to deploy, but it also makes them more exposed to unwanted security intrusions, as these sensors typically lack proper security protections, making Source Location Privacy a critical responsibility [9], [17]. Basically, source location privacy deals with safeguarding the location privacy of the endangered animals (or events/assets in general) from attackers and intruders.

To understand the importance of source location privacy (SLP) in WSN applications, let us consider a Panda-Hunter game model proposed in [18]. In this scenario, the pandas are thought to live in a monitoring area with a huge number of sensor nodes that are dispersed randomly and uniformly. When a panda is detected, the sensor node that detected the panda becomes the source and sends the monitored message in a multi-hop way to the Base Station (BS). However, the hunter/poacher whom we term as the attacker or the adversary is all set for poaching using smart ICT technologies. The attacker begins its journey from the BS location as the entire network traffic converges here. By studying the transmitted signal, the hunter can determine the position of the immediate sender node by *backtracking* the packets. The hunter repeats the procedure until it hits the source node and finally captures the panda. Therefore, it is important to conceal the position (i.e., ensure location privacy) of the node that identifies an asset in its radio range. To mitigate such issues, literature has several solutions for preserving SLP in WSNs. Each of these works have demonstrated solutions considering a different type of attacker models (local, semi-global and global attacker), network models (random, strategic, square, and circular networks), different ways to protect the assets (routing-, fake packets-, fake source-, multiple BS-based schemes). However, no work exists in the literature (to the best of our knowledge) to date, that have demonstrated the impact of the *network size* and the *network shape* on the performance effectiveness of the SLP techniques. To fill this research gap, we aim to explore this dimension i.e., study the effect of the size and the shape of the network on the performance metrics of the existing SLP solutions. Since different network models are used to produce different SLP solutions, it is critical to understand the impact of the shape and the size of the network models on the performance in order to make better use of the current SLP solutions. Particularly, we study and evaluate the effect of the two existing different network models, square and circular

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deployments, on the effectiveness of the performance metrics of SLP solutions proposed so far. Initially, the work begins with a description of the two network deployment models in detail. Followed by the description of the two existing random walk-based SLP techniques which we used in this work to study the behavior of the network size and shape effects of these two schemes. The following are the key contributions of this article are:

- Implement two different network models—square and circular.
- Study the impact of the shape and size of the two network models on the performance of the two existing SLP techniques namely, Phantom Routing Based Pure Random Walk (PRPRW) and Phantom Routing Based L-Path Pure Random walk (PRLPRW) in various network settings.
- Also, study the impact of the radio range of the sensors on the performance metrics of the existing SLP techniques.
- Provide inferences and findings of the work and decide which model is doing better in terms of performance metrics.
- Give future research directions to prospective researchers to explore hidden or unexplored issues in SLP designed for WSNs.

The rest of the paper is organized as follows: In sec. II, we give an overview of the existing SLP techniques. These SLP techniques have used either square or circular deployment models. In sec. III and sec. IV, we explain the network and attacker models considered in the work. We then describe the two SLP techniques used as case study protocols for this research in sec. V. The simulation setting, results, and discussions are presented in sec. VI, sec. VII and in sec. VIII respectively. Finally, we give conclusions and future work in sec. IX.

II. Literature Review

In this section, we present recurrently published SLP protocols proposed for WSN under the two broad categories. The categories of the solutions are made based on the shape (square or circular) of the network deployment considered in the work by the authors. Please refer to Fig. 1 and Fig. 2 in Sec. III for these two network models.

A. SLP Solutions with Square Deployment Model

The idea of a square deployment model for WSN is first seen in the works of Jhumka et al. in [19]–[22]. The work in [20] focuses on designing online parameter estimation dynamically for protecting source privacy in WSN. The primary focus is on understanding the trade-off between a temporary fake source period, a temporary fake source duration, and a permanent fake source period.

The objective of the work in [21] is to provide a decision-theoretic procedure for selecting an appropriate SLP routing algorithm from among several available ones. The choice is made based on the application's needs and the network under investigation.

A multi-objective optimization problem is suggested in [22]. In this work, SLP, schedule latency, and attacker distance are considered as main criteria as the optimization parameters. Based on these parameters and the use of generic algorithms, a scheduling algorithm is proposed for the generation of fake packets.

Three levels of security for SLP in WSNs are suggested in [23]. A ring surrounding the central BS serves as a fake packet flooding zone, a fake node that lies in the opposite direction to that of the source acts as a fake source and finally, an additional fake source in the appropriate quadrant of the network is all adding ambiguity to the network traffic. Thus, confusing the attacker.

The limitations such as poor data yield and low SLP on phantom routing was overcome in [24]. In the proposed phantom walkabouts a generalized routing in which phantom routes of variable lengths are suggested. The work has shown improved privacy.

By slicing packets into numerous fractions and transmitting through multiple dynamic paths, this paper [25] proposes a dynamic multipath privacy-preserving routing (DMPPR) strategy based on multiple sinks for source location privacy. The DMPPR strategy picks a phantom node at random for each transmission to avoid the nodes dying too soon. It generates multiple routing methods which disperse the routing distribution, resulting in increased energy usage in non-hotspot locations and closed-loop transmission is proposed as it reduces the likelihood of disclosing the source location in a short period of time thus enhancing the secure time and node utilization ratio.

In [26], the authors proposed two phantom routing-based methods for the scenario where multiple assets co-exist in the network. The first strategy namely, phantom routing based backward random walk (PRBRW), routes packets to the base station (BS) using a mix of backward random walk and a greedy forwarding approach. It performs better in terms of capture ratio and safety period, but it reduces network lifespan and has low entropy. A second strategy, called phantom routing based L-path random walk (PRLPRW), is suggested to overcome the drawbacks of the PRBRW techniques. It performs well in terms of metrics like capture ratio, safety period, and entropy. The main finding in this research is providing SLP for the scenarios with the network having multiple sources.

B. SLP Solutions with Circular Deployment Model

Circular deployment models in the domain of WSN were seen initially in the works by Tao et al. and Anfeng et al. in [27] and [28] respectively. Details of the circular deployment model are given in Sec. III. However, this work focus on energy-efficient routing techniques for WSNs with reliability feature enabled using secret sharing schemes. They do not focus on the SLP, unlike this work.

Inspired by the circular deployment models as in [27], [28], the authors in [29] (for the first time in the domain of SLP in WSN) have proposed a novel source location privacy solution. In this approach, the network is divided into circular annular rings with rings of the size sensor's radio range r_s. BS is positioned in the center of the network at which all the network traffic converges. Whenever the source sensor detected an event, it sends the packet using the proposed technique named 'SLP using Randomized Routes (SLP-R)'. In this solution, the packets are initially sent in the backward direction i.e., towards the network boundary for certain hops (TTL). Once the TTL value becomes zero, the packets are sent in either a

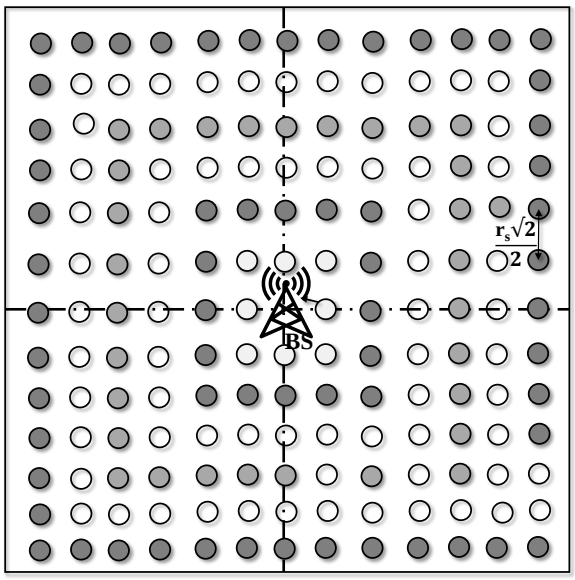


Fig. 1: Square Deployment Area

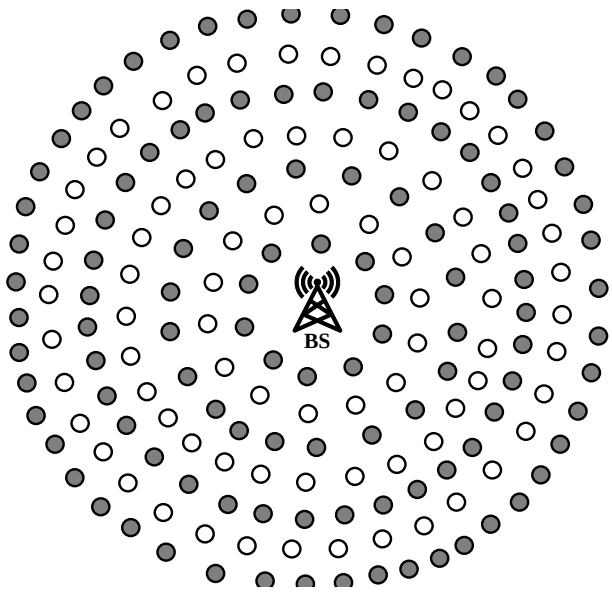


Fig. 2: Circular Deployment Area

clockwise or anticlockwise direction. This walk takes between $45°$ to $180°$. After this identical depth routing (IDR) phase, the packets are sent to the BS using shortest path routing (SPR). This solution has shown improved *safety period* and *network lifetime*.

The work in [30] involves sector-based random routing (SRR). The circular network is logically divided into several sectors instead of annular rings. It then employs a number of scattered routes in order to develop an enhanced SLP and optimize energy consumption. To get to the sink node in the SRR, the data packets are dispersed among phantom sources at random spread throughout several sectors. In order to protect the asset/event, the protocol ensures the selection probability of the nodes in those sectors that are near the source node is less. In order to maintain a balance between security and network lifetime, the idea of a hop threshold is also provided.

Another work that employs circular network topology for SLP is seen in [31]. In this model, in addition to annular rings, the nodes in the network are further grouped into clusters. Each node in the cluster forwards the information to the cluster head. The cluster head relays the packets to the BS via other intermediate cluster heads. In addition to regular routing, fake packets from fake sources are simulated to imitate the true behavior of the source node. This adds to uncertainty to the network traffic and increases the *entropy* of the system.

The aim of work in [9] is to develop an improved SLP preservation technique that gives a uniform privacy level for any position of the source/asset in the network or any network configurations. In this scenario, the circular deployment region is divided into annular rings and in addition, the network has three major annular segments each of size one-third the network radius. Based on the position of the source within these segments, the protocol adapts dynamically and performs four stages of data routing: i) Backward Random Walk (BWR), ii) Clock/anti-clockwise routing, iii) shortest path routing to the second ring, and iv) ring walk. These four phases are designed in such a way that privacy remains at a uniform level irrespective of the position of the source in the network.

Other works that employ circular deployment models are seen in [32]–[34].

III. Network Models

A network model consists of sensor nodes in which the node at the center acts as the base station (BS). All the information from the sensor nodes is transmitted to the BS. Each sensor node is composed of a transceiver to transmit the perceived data, sensors to scan the environment and identify or detect the targets, a microprocessor/microcontroller, storage/memory, and a power source. Additionally, we presume that the sensor nodes comprise an RFID reader and that the target asset(s) are fitted with an RFID transponder. Every time the asset enters the sensor node's range of detection, the RFID tag on the asset communicates with the reader on the sensor node. By using established SLP privacy protection methods, the sensor node then sends this data to the Base station. We assume that the sensor node's detecting range and communication range are both identical in our instance. All network communications are presumed to be secure and dependable. The nodes are homogeneous meaning that all nodes have the same properties such as energy capabilities and radio range (rs). The existing network models are broadly classified as square network models and circular network models, which are described in the following subsection.

A. Square Network model

As illustrated in Fig. 1, the network has a square deployment architecture with nodes arranged in a regular pattern and a uniform distance of $\sqrt{2}r_s/2$ units between any two adjacent nodes. At the core of the network is a single base station (BS) and the nodes are considered to be placed in square-shaped rings around the BS.

B. Circular Network Model

The circular deployment model is illustrated in Fig. 2 which includes strategically placed sensor nodes and a single central

base station (BS). The nodes are intended to be arranged in concentric rings around the BS as demonstrated. Each concentric ring is separated by a distance of r_s—radio range of the sensor node. Around the BS, the nodes are positioned angularly in the rings. The rings are arranged in a way that one made up of nodes that are only one hop distant from the base station is the closest. The second ring and third ring are made up of nodes that are two and three hops away from the BS, respectively. This process of node placement is extended to the edge of the network. As a result, the angularly oriented nodes come together to form a ring.

IV. Attacker Model

The local and passive attacker models are considered in this study. In this paradigm, the attacker just listens in on wireless communication links rather than launching active assaults like node compromise or packet content alteration. Furthermore, because the BS is the final destination of all packets, it is presumed that the attacker begins its backtracking effort there. That is, the adversary guesses the path of the packet's signal arrival for each new packet delivered from the source node to the BS. The attacker uses a device with a spectrum detecting and analysis module, as well as good amount of storage and computational capability, to determine the packet's arrival direction. According to this estimation, the attacker moves one hop in the direction of packet arrival. This shift is in the direction of the source of information and away from the BS. Furthermore, the attacker's devices are assumed to have the same sensing radius as the sensor nodes.

V. Routing Techniques

In this section, we shall briefly describe the working of the two existing routing protocols namely, Phantom Routing Phase Pure Random Walk (PRPRW) and Phantom Routing Based L-Path Pure Random Walk (PRLPRW) that shall be used as test cases or case study protocols to study the behavior of these SLP protocols with different network models, i.e; square and circle models.

A. Phantom Routing Based Pure Random Walk

PRPRW [18] works in two phases: i) the first one is the initialization phase during which each node in the network discovers its one-hop neighbor and its distance to the BS, and ii) the second one is the operation phase: in this phase, once the asset/event is detected by a sensor node it starts sending packets to the BS using the random walk method. Initially, the source chooses a neighbor node randomly from its neighbor list and relays the packet to that node. Before relaying the packet, the source node initializes the packet with time-to-live (TTL) parameter, which is a predefined hop count value. The neighbor node that receives the packet decrements this TTL value by one, chooses a neighbor randomly from its neighbor list, and relays this packet to that neighbor. This process repeats till the TTL value is non-zero. Once the TTL value is zero, the packet is relayed to the BS using the shortest path routing approach. The node that has a packet with a TTL value equal to zero is termed as a '*phantom node*'. The phantom node chooses a neighbor node that is closer to the BS compared to its other neighbors and relays the packets to that neighbor. Thus achieving the shortest path routing up to the BS.

B. Phantom Routing Based L-path Pure Random Walk

PRLPRW [26] works in three phases. During the first phase, the initialization of the network takes place. The working of this phase remains the same as in the PRPRW technique. In the second phase, the packet moves in the L-path direction from the source node. Initially, the packet is sent either vertically up or down for certain hops and then horizontally left or right for another set of hops. This phase is intended to drive the packets away from the source node by introducing biases in the ransom walk. Finally, in the third phase, the packets are sent to the BS using the shortest path routing approach. This technique has shown better results compared to the PRPRW scheme.

VI. Simulation Settings

500 packets from the source node are transmitted per simulation and a total of 50 simulations were carried out, using two protocols Phantom Routing Based Pure Random Walk (PRPRW) and Phantom Routing Based L Path Random Walk. These simulations are analyzed and evaluated in two different network topologies square and circle. The simulation starts when the source node sends a packet to the BS and stops when the attacker finds the source node or when the source node delivers all the packets. The network's size is kept constant throughout the simulations. In this setting, the source node's location in relation to the BS is changed. The source node is initially situated at a distance of 250 units away from the BS. Then, we gradually move the source node position by 250 units further after the first round of simulations, placing it at a 500 units distance from BS. This way the source node positions are located at 250, 500 and 750 units away from the BS. We used the Python programming tool to model the current work. To decide on the better deployment model, the simulations were carried out in four different combinations (or cases) described as follows:

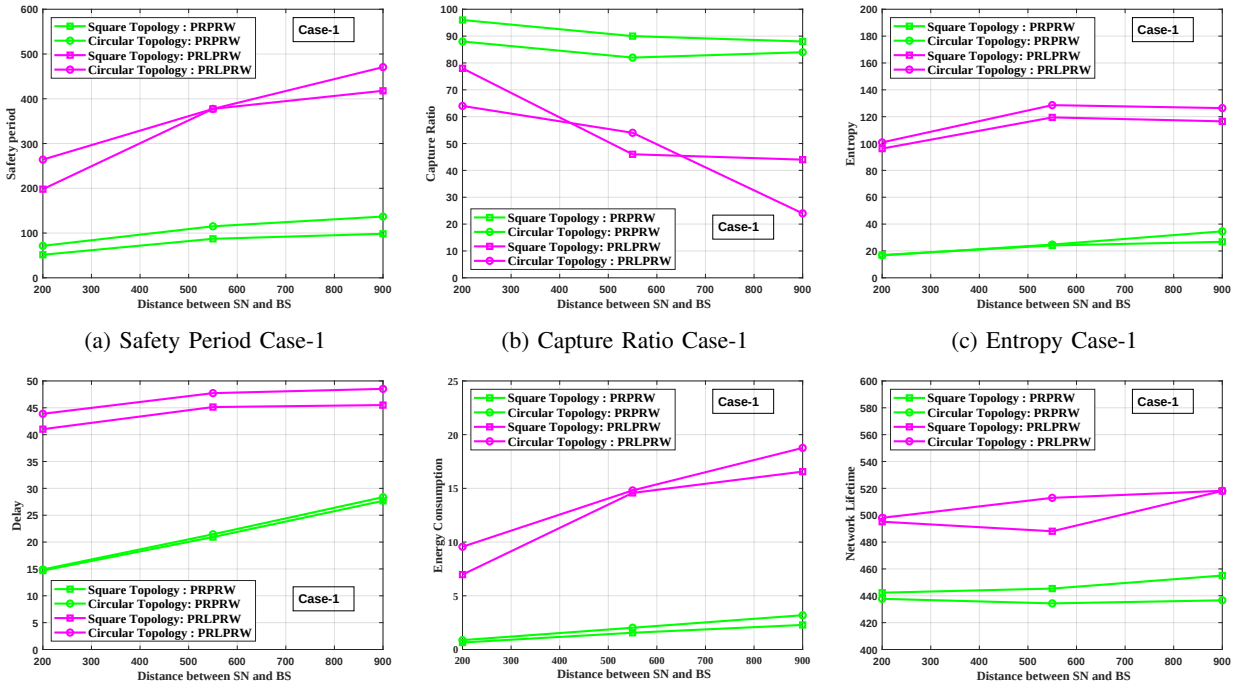
- Case-1: In this case, the network area and the sensors' radio range of both the network models (square and circular) were kept fixed; while the number of nodes were varied. An area of 450000 meters, 72 units of radio range, and an off-set distance of 50 units for both topologies were taken into consideration. The total number of nodes including the base station was 1849 and 1746 in square and circle network topologies respectively.
- Case-2: In this case, the number of nodes and sensors' radio range in both the topologies was kept fixe;, while the area spanned by the network was kept different. A total of 1369 nodes were taken in both the models. An area of 3422500 and 3141592 (values rounded off to nearest integer) square units was considered in square and circular models respectively.
- Case-3: This simulation setting is similar to the one suggested in Case-2. However, in this case, the radio range of the sensor nodes were varied. Radio ranges of 72, 77, 82, 87, 92, 97, 102, 107 and 112 values were set in the simulation in each trial.

A. Evaluation Metrics

Here, we list and briefly explain various metrics that have been considered in the literature to analyze and evaluatethe performanceof the SLP solutions.

TABLE I: Details of the network sizes and shapes

Cases	Area	No. of Nodes	Radio range r_s
1	Same (4500000 sq. m)	Different (Square: 1849 and Circular: 1746)	72 units
2	Different (Square: 3422500 and Circular: 3141593)	Same (1350 nodes)	72 units
3	Different (Square: 3422500 and Circular: 3141593)	Same (1350 nodes)	72, 77, 82, 87, 92, 97, 102, 107 and 112



(a) Safety Period Case-1 (b) Capture Ratio Case-1 (c) Entropy Case-1

(d) Delay Case-2 (e) Energy Consumption Case-1 (f) Network Lifetime Case-1

Fig. 3: Effect of square and circular deployment models on various performance metrics in PRPRW and PRLPRW schemes (Case-1)

1) Safety Period: The number of packets successfully delivered to the BS before the attacker captures the asset is known as the *safety period*.

2) Capture Ratio: The percentage of simulations in which the attacker successfully captures the asset before the simulation is complete is called the *capture ratio*, also referred to as the *capture percentage*. That is capture ratio indicates the *attacker's success rate*. We count the number of simulations in which the attacker was successful in capturing the asset before the simulation ended and estimate the percentage by measuring the total number of simulation trials.

3) Entropy: It is a measurement of the average level of uncertainty in a set of data. Later, the communication industry adopted this idea to determine network traffic uncertainty. The level of unpredictability of the routing patterns formed by random walks is measured using *entropy*. This gives the routing path randomness. The more random the paths are, the better the entropy is. This implies better is the safety of the asset.

4) Delay: This parameter allows us to determine how many hops data packets require to reach the BS from the SN, on average. Packet movement from one node to the next node is defined as a *hop*.

5) Energy Consumption: Processing information packets, as well as sending and receiving the packets, consumes energy. Only the energy expended during data transmission between nodes is taken into consideration in various literature.

6) Network Lifetime: It is defined as the time interval between the beginning of network operation and the first node's energy exhaustion. One approach employed in the literature to measure this metric is the number of packets sent to the BS till a node in the network dies. This definition holds true assuming that the network has only one asset i.e., one source of information.

VII. Results

In this section, we provide the main findings of the work. In particular, we try to understand the effect of network size (measured in terms of the number of nodes and total area of the network) and shape (square and circular deployment models) on the performance of the existing SLP solutions (used as test cases).

Here, we describe the performance metrics for the case in which the simulation scenario consisted of the same area in both the square and circular topologies but the number of sensor nodes was different. For details, please refer to Tab. I. We label this scenario as Case-1 and use this label as an identifier to throughout the remainder of the paper.

The safety period metric is shown in Fig. 3a. The green-colored lines indicate the plots for the PRPRW protocol and the magenta colored lines indicate the plots for the PRLPRW protocol. It is observed that the safety period for the circular

deployment model is better when compared to the safety period of the square deployment model. This is true for all the distances between the SN and the BS. While the safety period of the PRLPRW technique is better than the PRPRW technique, we find that even here the circular topology is doing better in terms of safety period metric when compared with square topology. This is true for the scenarios when the SN is closer to the BS and at the network edge. However, the safety period in the case of PRLPRW is the same for both the square and circular topologies. The inference is that the circular deployment model is doing better in terms of safety period metric, on average, in both protocols.

The plots in Fig. 3b are for the capture ratio metric. Its value, in the PRPRW technique, is higher in the square deployment model when compared with the circular deployment topology. This holds true when the source is situated closer to the BS and when near the network edge. However, its value is lower than that of the circular deployment model when the SN is in the middle of the network. Coming to the PRLPRW solution, the capture ratio has lower values in circular deployment when compared with the square deployment model. This is true for all the distances between the SN and the BS. The lower the capture ratio is, the better the privacy of the asset. On average, in either case, the circular deployment topology is performing better than the square deployment topology.

Next, we discuss about the entropy metric. The plots for the entropy metric are shown in Fig. 3c. It is observed from the plot that the values of the entropy metric for PRPRW are almost the same for both the circular and square deployment models. This is true for distances when SN is closer to the BS and in the middle of the network. However, when the SN is situated near the network edge, the circular deployment model has slightly better entropy when compared with square topology. It is seen that the amount of entropy for circular topology is better than in square topology, for all the distances between the SN and the BS.

The plots for the delay metric are seen in Fig. 3d. The amount of delay (measured in hops) is almost the same for both deployment topologies in the case of the PRPRW protocol. This is shown in green colored lines. However, in the PRLPRW protocol, the circular deployment model has a higher delay when compared with the square deployment model. This is true for all the distances between the SN and the BS. However, it is should be noted that better privacy is achieved at the cost of increased delay.

Energy consumption is another metric often taken into consideration while designing any type of routing protocol. From Fig. 3e, it is observed that for smaller distances between the SN and the BS, in the PRPRW technique, circular and square topologies exhibit the same about of energy expenditure for sending information packets to the BS. However, as the distance between the SN and the BS increases, the circular deployment model requires higher energy to deliver the information packets to the BS when compared with the square deployment model's energy expenditure. Similarly, for the PRLPRW scheme, circular topology needs more energy to drive information packets to the BS. This holds true for the cases when the SN is closer to the BS and near the network edge. However, the amount of energy expended in driving the packets to the BS when the SN is situated in the middle of the network remains the same in both deployment models. This plot indicates that the square deployment model is doing better in both protocols when compared with the circular deployment model.

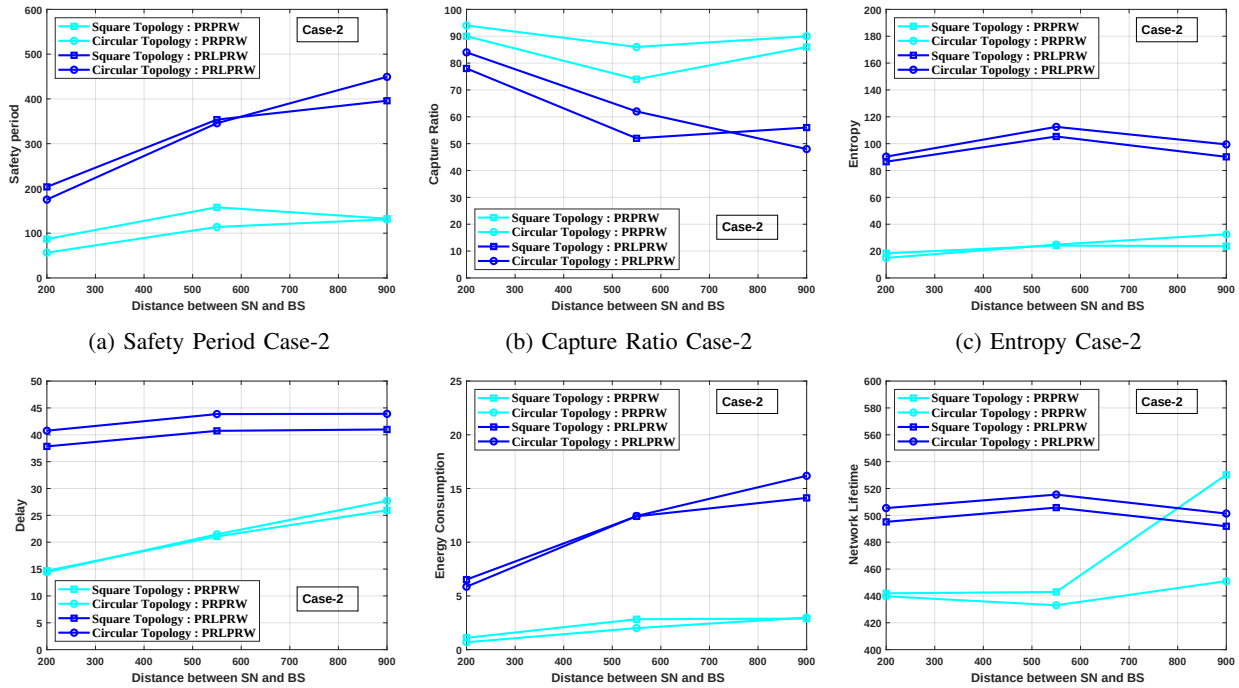
Another interesting metric often used in WSNs is network lifetime (NLT). This metric indicates the active lifespan of the network since its deployment time. Referring to Fig. 3f, the NLT metric has almost same the magnitude for both the square and the circular topologies when the SN is closer to the BS. However, as the distance between the SN and the BS increase, square topology is doing better in terms of NLT when compared with the circular deployment topology in the case of the PRPRW technique. This is shown in green colored lines in the figure. Coming to the PRLPRW scheme, the NLT in both topologies is the same when the SN is near the BS as well as near to the network edge. However, its magnitude, in the circular deployment model, is larger when compared with the NLT magnitude in the square deployment model. This is shown in magenta-colored lines in the figure. The inference drawn from this metric is that square topology is doing better, on average, in PRPRW technique. Circular deployment model is doing better in PRLPRW technique.

Next, we describe the performance metrics for the case in which the simulation scenario consisted of the approximately same number of sensor nodes in both the square and circular topologies but the areas of the deployment regions were different. For details, please refer to Tab. I. We label this scenario as Case-2 and use this label as an identifier throughout the remainder of the paper.

The safety period metric for Case-2 is shown in Fig. 4a. The cyan colored lines are the values for the PRPRW protocol and blue colored lines are the values for the PRLPRW protocol. For smaller distances between the SN and the BS, the values of the safety period in the circular deployment model are low compared with the values of the safety period in square topology. This trend holds true for distances between the SN and the BS from the low range to the mid-range. However, as the source position moves closer to the network, the values of the safety period remain the same in both the square and circular deployment models. In the case of the PRLPRW protocol, the square deployment model is doing better in terms of safety period for smaller distances between the SN and the BS. This trend, however, reverses when the distance between the SN and the BS increases. Thus, when the source is situated at the network border, the safety period metric in the circular deployment model is better when compared with the values of the safety period in the square deployment model.

The capture ratio metric for both the protocols PRPRW and PRLPRW is shown in Fig. 4b. For low and mid-range distances between the SN and the BS, the success rate of the attacker is lower in the square deployment model when compared with the circular deployment model. Only for the larger distances between the SN and the BS, the circular deployment is doing better than the square deployment model. On average, the square deployment model is doing better than the circular deployment model in the PRPRW protocol.

Routing path uncertainty is usually measured in terms of entropy. Fig. 4c shows this metric for both the protocols and for both the deployment models. It is seen that the values of the entropy metric are almost the same for both the deployment



(a) Safety Period Case-2 (b) Capture Ratio Case-2 (c) Entropy Case-2

(d) Delay Case-2 (e) Energy Consumption Case-2 (f) Network Lifetime Case-2

Fig. 4: Effect of square and circular deployment models on various performance metrics in PRPRW and PRLPRW schemes (Case-2)

models, in the PRPRW protocol, when the source is closer to the BS. This is also true when the source is located in the middle of the network. However, the entropy of the circular deployment is better than the square deployment model when the source is at the network edge. In the case of the PRLPRW protocol, the circular deployment model is doing better for all the distances between the SN and BS than the square deployment model.

Fig. 4d shows the delay metric for the square and circular deployment models for case-2 in both the PRPRW and PRLPRW protocols. The values of the delay metric are almost the same for low and mid-range distances between the SN and the BS in the PRPRW technique in both square and circular deployment models. However, when the source is closer to the network edge, circular deployment has slightly more delay when compared with the square deployment model. Coming to the PRLPRW protocol, the circular deployment model has a higher delay when compared to the square deployment model for all the distances between the SN and the BS.

The energy consumption metric for case-2 is shown in Fig. 4e for both the protocols and for both the topologies. The energy expenditure for the case of square and circular deployment models in PRLPRW has almost the same values for all the distances between the SN and the BS. A similar trend is seen in the PRLPRW technique for both deployment models. This trend holds true for low and mid-range distances between the SN and the BS. However, for longer distances, the circular deployment model has larger energy consumption when compared to the square deployment model.

Finally, coming to the NLT in case-2, PRPRW exhibits the same about of NLT when the source is closer to the BS in both circular and square models. However, as the distances between the SN and the BS increase, the NLT in the square deployment model is doing better when compared with the circular deployment model. In case of the PRLPRW technique, the circular deployment model has better NLT for any distances between the SN and the BN.

VIII. Discussion and Analyses

Although, in some scenarios square deployment model is showing better performance, in the majority of the cases, circular deployment has performed better on an average. This leads to the conclusion that circular deployment is better than the square deployment model. To further support this inference, we have computed the average of all the values for of these each metrics in both the protocols; for both the deployment models. We then computed percentage deviation to understand which topology is doing better. In particular, the percentage deviation is computed as the ratio of the difference in metric values of circular topology and the square topology to the values in the square deployment models. These computed deviations (increased/decreased values) for both the protocols PRPRW and PRLRW are shown in Tab. II and Tab. III respectively.

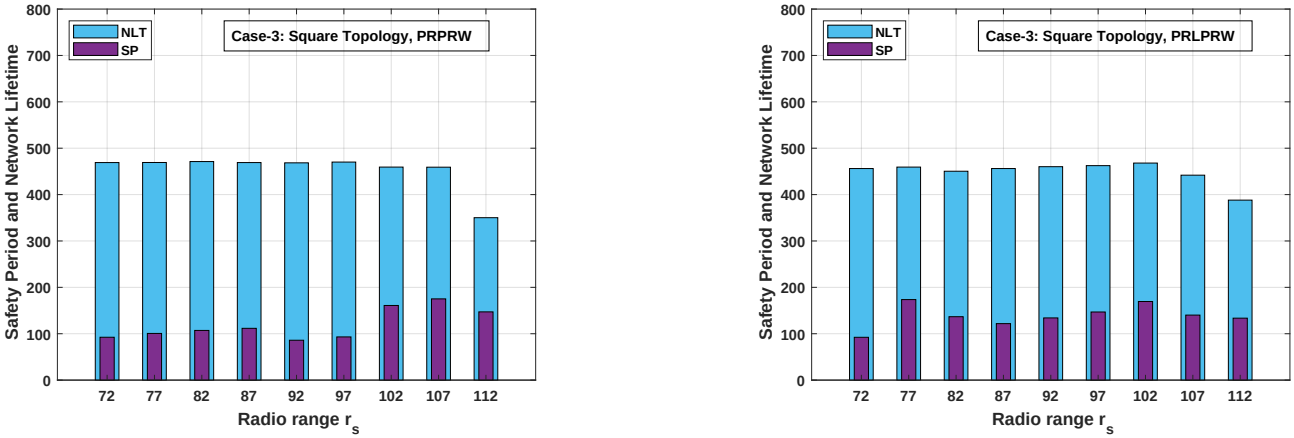
Tab. II shows the percentage deviation of both the protocols PRPRW and PRLPRW for the scenario labeled case-1. It is clear that all the performance metrics except for the NLT have shown better results in the circular deployment model compared to square one. However, the NLT metric is doing better in the square deployment model when compared with the NLT in the circular deployment model. NOTE: The capture ratio metric will have negative values as this indicates the attacker's success rate. The larger the negative magnitude of this metric, the lesser the attacker's success rate. Similarly,

TABLE II: Percentage deviation of performance metrics in PRPRW and PRLPRW for Case-1

Metrics	% Deviation in PRPRW	% Deviation in PRLPRW
Safety Period	36.64	15.32
Capture Ratio	-07.25	-15.33
Entropy	10.02	7.87
Delay	01.9	6.36
Energy Consumption	34.44	17.35
Network Lifetime	-02.54	01.89

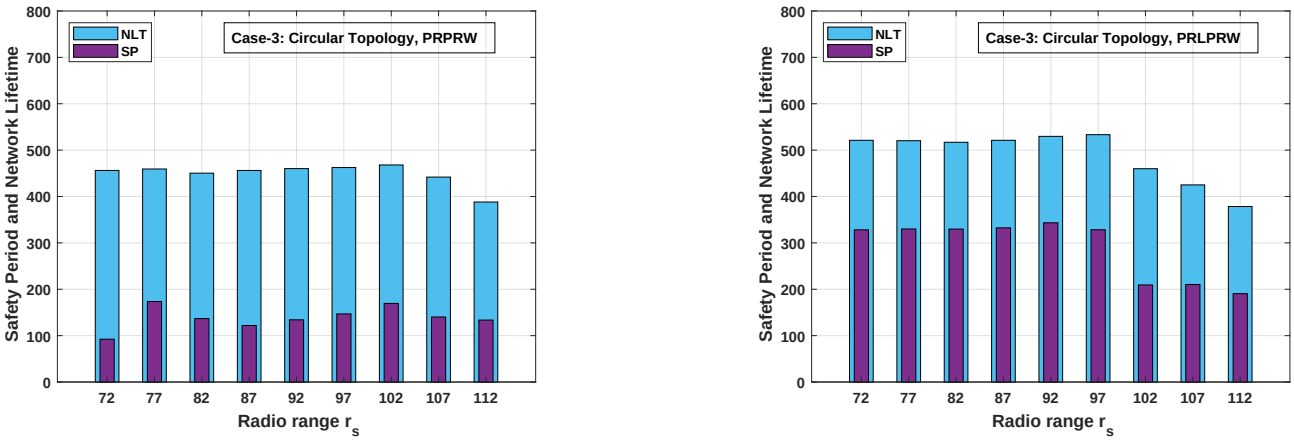
TABLE III: Percentage deviation of performance metrics in PRPW and PRLPRW for Case-2

Metrics	% Deviation in PRPRW	% Deviation in PRLPRW
Safety Period	-21.3	-0.91
Capture Ratio	08.43	07.14
Entropy	07.16	7.87
Delay	02.29	07.49
Energy Consumption	-21.16	01.53
Network Lifetime	-05.89	01.96



(a) (b)

Fig. 5: Safety Period and Network Lifetime vs Radio Range. Case3: (a) Square, PRPRW and (b) Square, PRLPRW



(a) (b)

Fig. 6: Safety Period and Network Lifetime vs Radio Range. Case3: (a) Circular, PRPRW and (b) Circular, PRLPRW

Tab. III shows the percentage deviation for the scenario labeled as case-2 in both PRPRW and PRLPRW protocols. In this case safety period metric and capture ratio (positive values) are doing better in the square deployment model when compared with the safety period of the circular deployment model. The other metrics namely, entropy and delay, are doing better in circular deployment than in square topology. Coming to energy consumption and NLT metrics, square topology is doing better in PRPRW and circular topology is doing better in the PRLPRW solution. Hence, on an average, circular topology is doing better than the square deployment model in the majority of the cases.

A. *Impact of Sensors' Radio Range r_s on Performance Metrics*

For the first time, to the best of our knowledge, we aimed to study the impact of sensors' radio range r_s on *safety period* and *NLT* in both the protocols and in both the deployment models. Our long-awaited question "will radio range have any impact on strength of the privacy in SLP?" is answered in this article. To study this relationship between the nodes' radio range on safety period and NLT metrics, we used the same simulation settings that were used in case 2. But this time the sensors' radio range was incremented by 5 units in each simulation setting starting from the radio range of 72 units and ending at 112 units. A total of 9 settings were used to study

this dimension. The results obtained were thus averaged and the values are plotted in Fig. 5a to Fig. 6b. These four figures are for the square and circular deployment models in both the PRPRW and PRLPRW techniques. We label this scenario as case-3 and use this name in the rest of the paper.

Fig. 5a shows the bar plots for case-3, square topology, and PRPRW protocol. This figure shows both, the safety period and NLT, metrics drawn overlapped. The blue colored bars are for NLT and the purple colored bars are for the safety period metric. As the radio range increases the magnitude of the safety period increases. This trend is followed for radio range values of 72 to 87. Then at 92, the magnitude of the safety period metric falls lower. However, from there the trend is again increasing up to 107 units of the radio range. Then again it drops at 112. That is the increase in safety period with the increase in radio range holds true for certain values of r_s, then decreases in the next step and then follows the increasing trend. The safety period metric follows a zig-zag trend with an increase in radio range. However, this trend does not hold true for the NLT metric. The magnitude of the NLT remains almost the same for the values of the radio range between 72 and 97. From thereafter, NLT sees a decreasing trend in magnitude with an increase in radio range (i.e., for r_s values of 102, 107, and 112).

The plots for the safety period and NLT metrics vs. radio range for square topology in PRLPRW is shown in Fig. 5b. The strength of the safety period increases with an increase in radio range, then it decreases at one point, and from here on it gradually increases. This trend is seen in the figure. We see that for the radio range value of 71 and 77 there is an increasing trend, at 82 and 87 it drops by a small portion. Then from 87 onwards, the value of the safety period increases with an increase in radio range up to 102 units. Then again at 107, there is a dip in its value. That is an increase is radio range increases safety period but then this trend is for certain settings of the radio range values. Now, coming to the impact of radio range on NLT, we observe from the figure that radio ranges of 72 and 77 have the highest values of NLT. For radio range values of 82, 87, 92, 97, and 102, we obverse a gradual increasing trend in NLT, and finally the peak values match that of the NLT at 77. For the remaining cases of the radio range, i.e., for 107 and 112, we find a decreasing trend with an increase in radio range.

An observation to be made here is that the magnitude of the safety period of the square deployment model in PRLPRW is more when compared with that of the safety period in PRPRW protocol for the same square deployment model.

Fib. 6a and Fig. 6b shows the relation between safety period and NLT for different values of the radio range in PRPRW and PRLPRW techniques respectively, for both the circular topology in case-3. Fib. 6a shows that as the radio range increases, in the PRPRW scheme, the safety period metric has an increasing and decreasing trend. However, the values of the NLT metric remain almost the same for radio ranges from 72 to 102 units. For the radio range of 107 and 102, it exhibits a decreasing trend.

Fig. 6b shows that the values of safety period metric in the PRLPRW scheme remain almost same for the radio range value from 72 units to 97 units and from here after, its magnitude decreases. Similarly, the figure shows that the NLT remain almost at the same level for radio ranges from 72 to 97 units and for the rest of the radio range values, the NLT exhibits a decreasing trend.

Another observation to be made here is that the magnitude of both the safety period and NLT are higher in the PRLPRW scheme when compared with the values of the safety period and NLT in the PRPRW scheme. Because, PRLPRW [26] basically performs better than PRPRW solution [18].

B. Future work

We have been wondering if the network shape and size as well as the effect of sensor nodes' radio range have any effect on the performance of the existing SLP protocols. This paper is one such means to explore this direction and fill this research gap. Although there are quite a good number of research articles on SLP in WSNs, this preliminary study in the desired direction has focused only on one class of SLP techniques and that is random walk-based SLP solutions. Other categories of solutions that employ fake source or fake packet concepts to provide SLP are been not considered in the study as it is felt that WSNs require more energy to simulate these features. Since energy is one of the prime concerns, our focus at present has been only on random walk-based SLP solutions. So, one possible research direction for prospective researchers is to study the behavior of fake sources or fake packets based SLP for different shapes and sizes of the network model as well the impact of radio range on the performance effectiveness. Also, this initial research has focused only on strategically placed sensor deployment models, be it square or circular, and has not focused on random node deployment models. These models are also realistic in nature and in real-life scenarios. Another direction is to study the said topic using random deployment models and compare them with strategic deployment ones. Even though we understand that based on the application scenarios different network models are used, it is felt that an understanding of the knowledge of size, shape, radio ranges, and deployment of the sensor network will help the team or engineers while building their own networks for that particular application. In this preliminary investigation, we have focused only on two protocols namely PRPRW, the pioneering work in SLP, and PRLPRW techniques. It is desirable to test this suggested issue (impact of size and shape of network on SLP) on other protocols to make a better decision on the choice of usage of the network shape and size. Owing to the page restriction, we have restricted ourselves to only these two protocols. An interesting direction to be explored is the relationship between the coverage and the connectivity on performance metrics in source location privacy for WSNs. To the best of our knowledge, no work exists in this direction to date. Finally, an optimization problem involving the number of sensor nodes and the sensor's radio range values have to be explored to get the optimum performance of the SLP schemes.

IX. Conclusion and Future Work

Source location privacy in WSN has been the topic of the decade. Various solutions have been proposed in the literature considering various attacker models and various deployment models. However, the choice of the right network deployment models to achieve better privacy in WSN has not received attention so far. To fill this gap, this article has studied the

impact of the network size, and network shape as well the influence of the radio range on the performance of the existing SLP techniques. Two deployment models namely, square and circular models were studied in this work to see their impact on the performance effectiveness of two SLP techniques PRPRW and PRLPRW. The article has provided research findings for the scenarios of fixed network areas with variable numbers of nodes in both square and circular network topologies; and vice versa. It tries to answer the question " will network shape and size matter for SLP in WSNs?". In addition, the work also explores the impact of radio range on privacy and network lifetime for different network models. The study reveals that the circular deployment model has better performance than the square deployment in the majority of the cases. It also explored the influence of radio range on safety period and network lifetime and concludes that an optimal value of radio range has to be chosen to have better performance metrics. Finally, the work concludes by providing future research directions.

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